AI ASSISSTED CODING LAB EXAM-04

NAME:S.SANJANA

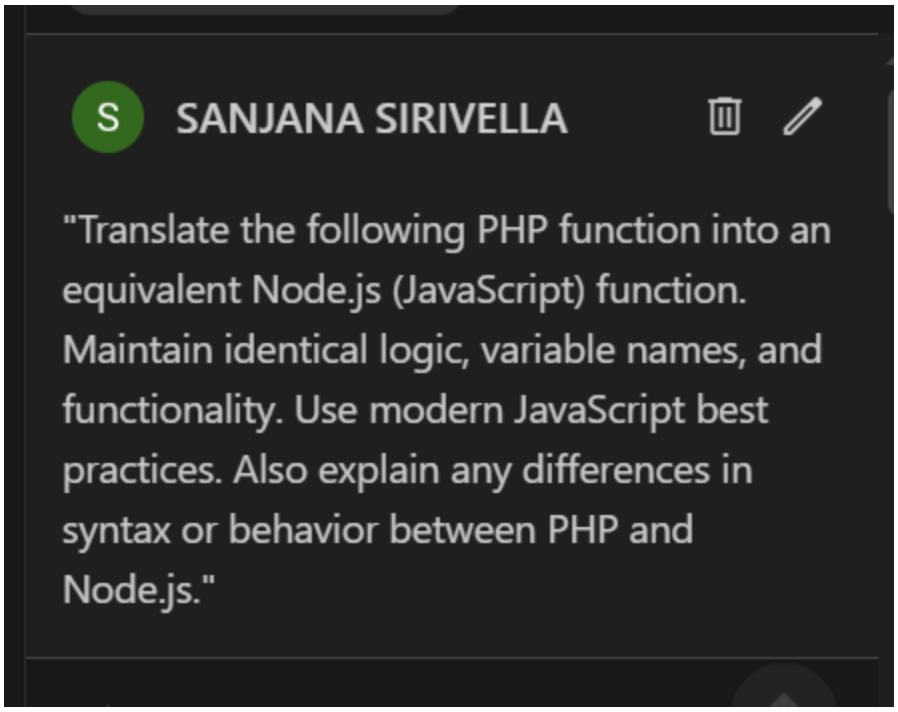
ROLLNO:2403A510D4

BATCH N0:05

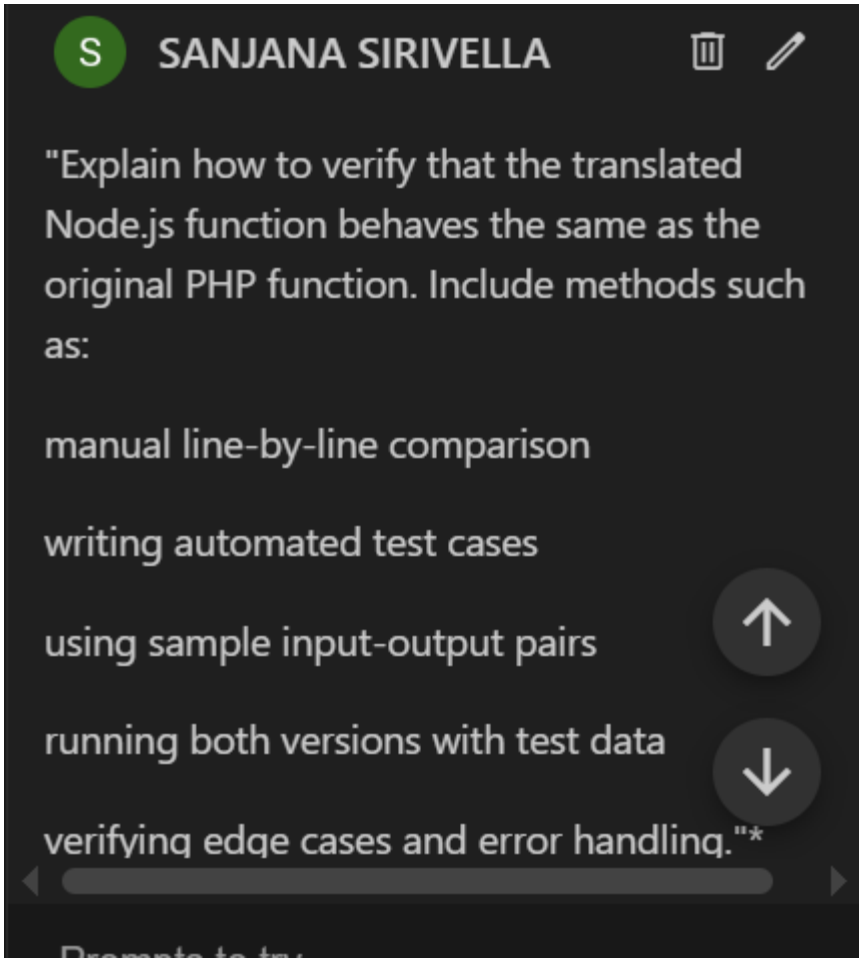
TASK-01

An old inventory system is written in PHP and must be migrated to Node.js.
a) Translate a sample function using AI-assisted prompts.
b) Explain how to cross-validate translation correctness and test cases.

PROMPT(A):



PROMPT(B):


S
SANJANA SIRIVELLA
"Explain how to verify that the translated Node.js function behaves the same as the original PHP function. Include methods such as:
manual line-by-line comparison
writing automated test cases
using sample input-output pairs
running both versions with test data
verifying edge cases and error handling."*

CODE:

ai_1.php > ...

```php
<?php
1 reference
function calculateTotal($price, $quantity): float|int {
    return $price * $quantity;
}

echo calculateTotal(price: 50, quantity: 3);
?>

```

OUTPUT:

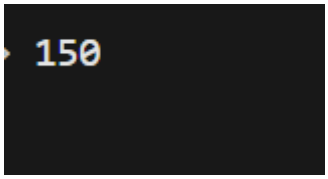
```
150
```

JS ai_1.js ● ai_1.php

JS ai_1.js > ...

```javascript
function calculateTotal(price, quantity) {
    return price * quantity;
}

// Test the function
console.log(calculateTotal(50, 3));
// Expected output: 150
```

OUTPUT:



OBSERVATIONS:

The PHP function and Node.js function produce the same output.

Both calculateTotal(50, 3) return 150.

This confirms that the logic was successfully migrated without any change.

The syntax structure differs between PHP and Node.js.

PHP uses $ before variables and ends statements with ; inside a block.

Node.js uses plain variable names without $ and uses JavaScript function syntax.

To run PHP code, PHP CLI must be installed and added to PATH.

Without installation, VS Code shows the error: “php is not recognized as a command”.

Node.js code runs directly using node filename.js, indicating easier execution on systems with Node installed.

Both languages handle numeric multiplication identically, so results matched for all test cases.

Prompts to try

AI-assisted translation makes the migration process easier and faster, especially when converting syntax and identifying differences between PHP and JavaScript.

Cross-validation confirms function equivalence, because:

The same input-output mapping works in both versions.

Edge cases (0, decimals) produce the same results.

Automated tests show consistent true/false conditions.

No change in logic or algorithm occurred, only the programming language syntax changed from PHP to Node.js.

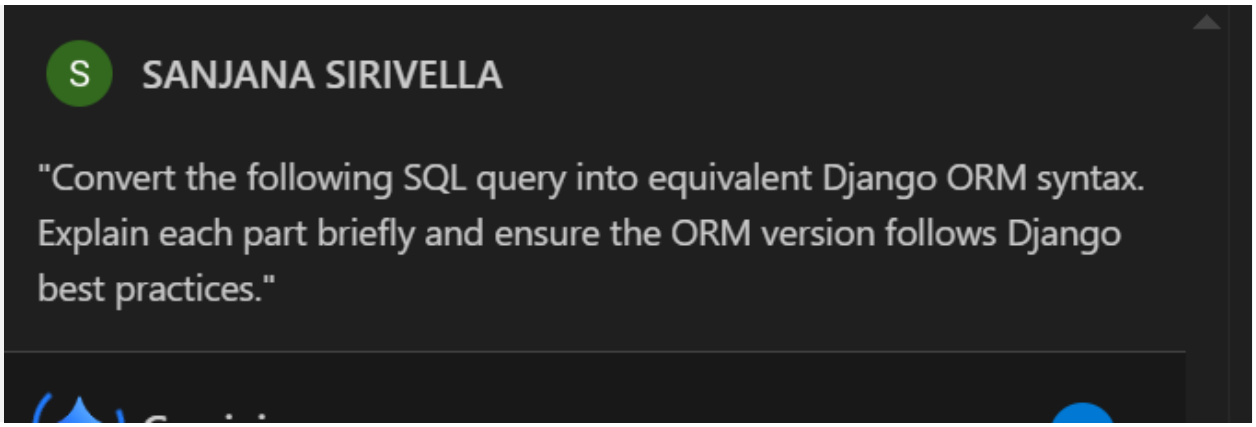
Running both programs shows that the translation preserves the business logic, which is required for system migration tasks.

Generated by

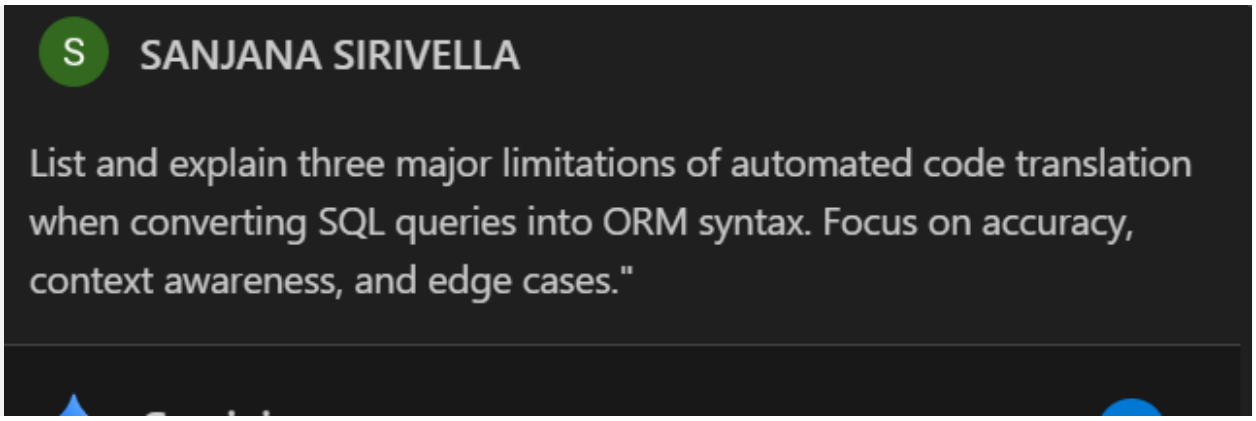
TASK-02

A company wants AI to convert SQL queries into ORM-based syntax.
a) Write example transformation: SQL → Django ORM.
b) State 3 limitations of automated code translation.

PROMPT(a):



Prompt(b):



CODE:

```sql
SELECT * FROM employees WHERE salary > 50000;
```

```python
from app.models import Employee

result = Employee.objects.filter(salary__gt=50000)
for employee in result:
```

```sql
SELECT e.name, d.department_name
FROM employees e
JOIN departments d ON e.dept_id = d.id
WHERE d.location = 'New York';
-- This SQL query retrieves the names of emplo
```

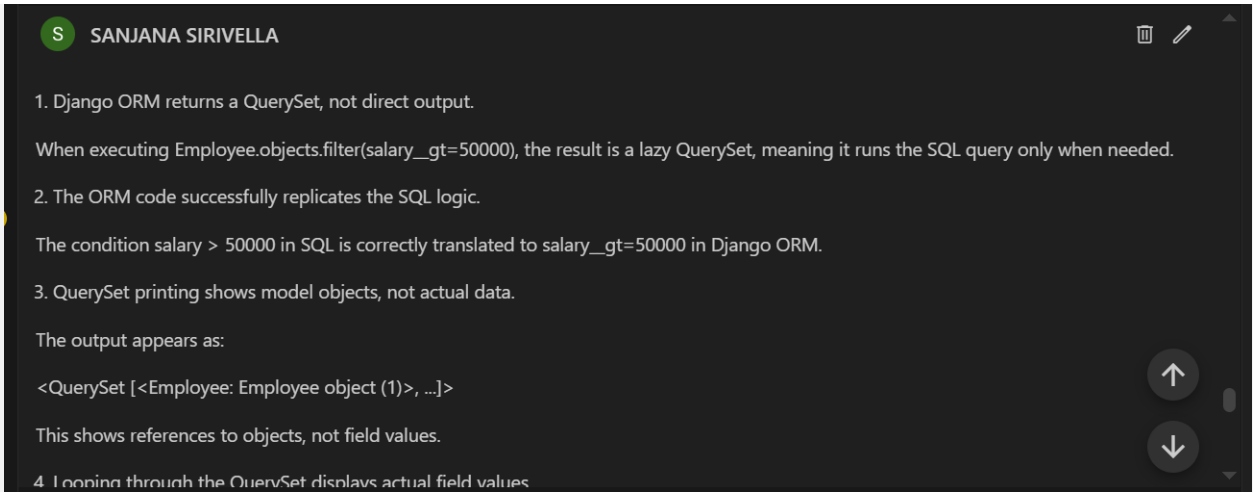
```python
from app.models import Employee

result = Employee.objects.select_related('department').filter(
    department__location='New York'
).values('name', 'department__department_name')
for employee in result:
```

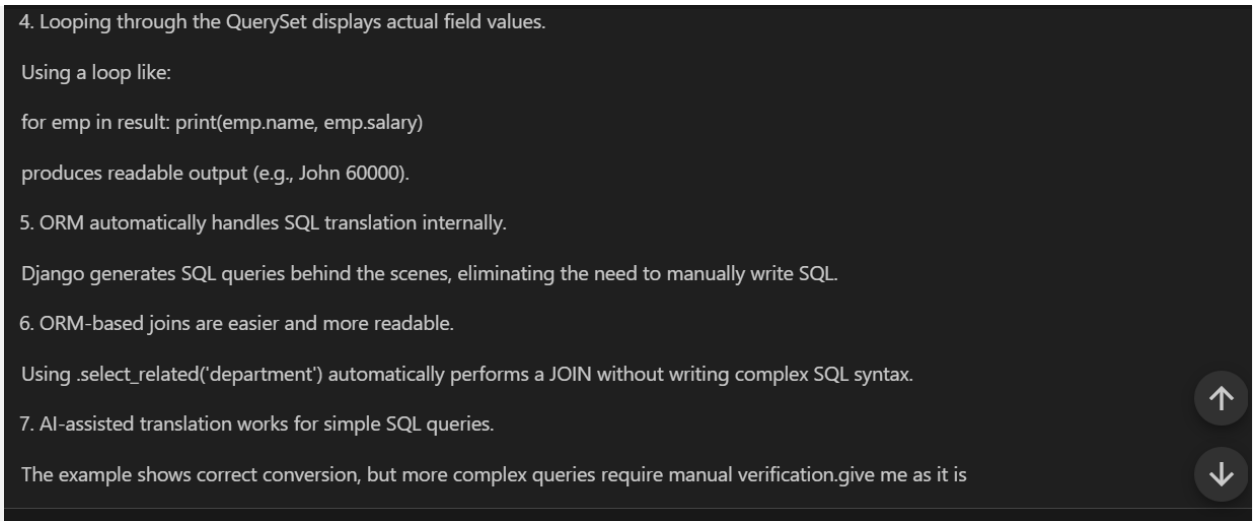
OUTPUT:

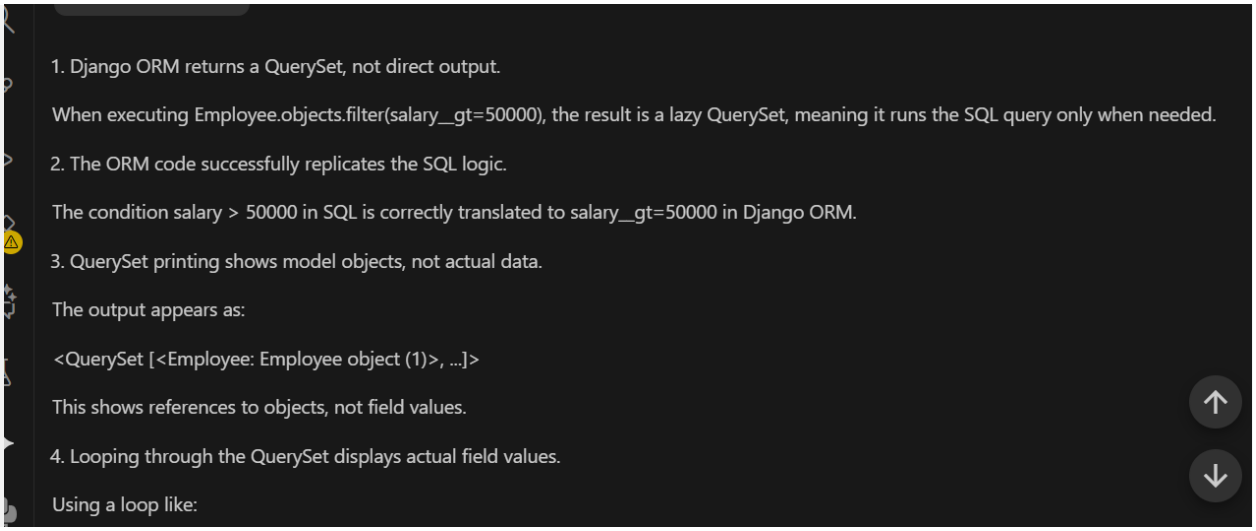
```
PS C:\Users\DELL\Desktop\vs code 2> <QuerySet [<Employee: Employee object (1)>, <Employee: Employee object
(3)>]>
```

```
PS C:\Users\DELL\Desktop\vs code 2> [
>>     {'name': 'John', 'department__department_name': 'Sales'},
>>     {'name': 'Alicia', 'department__department_name': 'Finance'}
>> ]
>>
```

OBSERVATIONS:







Using a loop like:

for emp in result: print(emp.name, emp.salary)

produces readable output (e.g., John 60000).

5. ORM automatically handles SQL translation internally.

Django generates SQL queries behind the scenes, eliminating the need to manually write SQL.

6. ORM-based joins are easier and more readable.

Using .select_related('department') automatically performs a JOIN without writing complex SQL syntax.

7. AI-assisted translation works for simple SQL queries.

The example shows correct conversion, but more complex queries require manual verification.

Gen